10.1 Introduction

In this lesson, we have to learn about the periodical maintenance procedure for our house holding electrical equipments, motors and electrical machines in industries. Also we have to learn briefly about the reasons for the faults, rectification methods and avoiding methods.

10.2 Electrical machines maintenance

10.2.1 Aim of maintenance

It is essential to maintain the electrical machines and their secondary instruments in our houses, workshops and industries for their proper operation. If we maintain properly there will not be any longer from the equipment's and they will function for the long period with high efficiency.

10.2.2 Maintenance

Maintenance is the process of operation needed for the functioning of the motors and other electrical machines with good efficiency. When the electrical machines work, they deliver mechanical power from it. So, it causes to worn out in their rotating parts. Hence the proper maintenance should be needed for the electrical machines.

10.2.3 Planned maintenance

The structure, operating methods, load level of the electrical machines, can be maintained daily, weekly, monthly and yearly. This process of inspection and proper maintenance work is called as planned maintenance.

10.2.4 Types of maintenance

i. Routine maintenance
ii. Preventive maintenance
iii. Break down maintenance
iv. Capital repairs or corrective maintenance

i. Routine maintenance

Based on the electric machines type, age and workload, the process such as cleaning, greasing, minor repairing works and minor adjustments done are called routine maintenance.

ii. Preventive maintenance

To prevent the sudden failure of the machines and to protect from major repairs, the planning of preventive maintenance should be followed. It will reduce the losses.

If the electric machine undergoes sudden breakdown, then two types of losses will occur.

a. Direct losses

It is defined as the repair charges for electric machines.

b. Indirect losses

It is based on the worth of loss of work, labour salary and unable to supply the production materials in particular time.

iii. Breakdown maintenance

If breakdown occurs suddenly after doing the routine maintenance and preventive maintenance, the machine is to be immediately repaired to a good condition is called breakdown maintenance.

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